

Zhuoli Ouyang

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Education

Southern University of Science and Technology

Undergraduate, Information Engineering

GPA 3.88/4.00 Rank 6/48

Shenzhen, China

2023 to 2027 (expected)

- Fields Elite Class (Honors), Department of Mathematics
- Honors Class, Department of Electronic & Electrical Engineering

2023 to 2024

2024 to Present

Dartmouth College

Visiting Student

Advisor Yaoqing Yang

Hanover, NH

Mar 2025 to Sep 2025

Westlake University

Visiting Student

Advisor Tailin Wu

Hangzhou, China

Oct 2025 to Aug 2026

Research Experience

ENGINE Endogenous Variational MultiScale Optimization for Zeroth-Order LLM

NeurIPS 2026

Fine-Tuning

Accept (Poster)

Zhuoli Ouyang (Project Leader), Changxi Chi, Siyuan Li, Tailin Wu

Advisor: Prof. Tailin Wu

Abstract: Existing zeroth-order optimization (ZOO) methods for LLM fine-tuning face a trade-off between gradient-estimation variance and curvature information. We propose ENGINE, a memory-efficient ZOO framework for LLM fine-tuning. Inspired by Variational Multiscale (VMS) in computational mechanics, we treat the LoRA space as coarse and the full parameter space as fine. A V-cycle couples the two scales, using full-space curvature feedback to correct low-dimensional updates. We develop a theoretical analysis explaining how ENGINE reduces both variance and bias. ENGINE achieves SOTA performance among ZOO methods on diverse LLM fine-tuning benchmarks, with faster convergence.

My Contributions:

- Formulated ENGINE by extending Variational Multiscale (VMS) theory from computational mechanics to ZOO. Developed the theoretical analysis explaining how coarse & fine coupling reduces both variance and bias.
- Designed the V-cycle algorithm with bidirectional state transfer based on the theoretical derivations. Independently implemented ENGINE, baselines, and the evaluation framework. Organized the complete codebase.
- Conducted the literature review. Designed experiments across model scales and downstream tasks. Developed the paper's narrative and structure. Wrote the full manuscript and handled the entire rebuttal process.

RMNP Row-Momentum Normalized Preconditioning for Scalable Matrix-Based

ICML 2026

Optimization [Paper] [Project] [Code]

Accept (Regular)

Shenyang Deng*, Zhuoli Ouyang*, Tianyu Pang, et al. *Equal contribution

Advisor: Prof. Yaoqing Yang

Abstract: Muon is widely used for LLM pretraining, but its Newton-Schulz iterations incur substantial computational overhead. We propose RMNP, an efficient row-normalized optimizer motivated by prior findings on block-diagonal dominance in Transformer Hessians. Our observations further support diagonal dominance in momentum Gram matrices. Under this assumption, we derive RMNP's preconditioner as a diagonal approximation to Muon's. The two become asymptotically equivalent in the diagonal limit. RMNP replaces Newton-Schulz iterations with row-wise momentum normalization, reducing preconditioning complexity to $O(mn)$. It substantially reduces preconditioning time while maintaining competitive pretraining performance with Muon.

My Contributions:

- Implemented and organized the complete RMNP, baseline, and benchmark codebase.
- Designed and ran all experiments, including benchmark comparisons and systematic ablation studies.
- Analyzed and organized experimental results. Prepared figures, tables, and experimental sections of the paper.
- Participated throughout the rebuttal. Prepared reviewer responses grounded in experimental and theoretical analysis.

NOVA Nullspace Dynamics in Neural Basis Solvers for Moving Boundary PDEs, Transport, Rank Transitions, and Reprojection

Submitted to ICLR 2027

Zhuoli Ouyang (Project Leader), Changxi Chi, Siyuan Li, Tailin Wu

Advisor: Prof. Tailin Wu

Abstract: Enforcing moving boundary constraints is challenging in multiphysics coupling. We propose NOVA to substantially reduce boundary errors while maintaining high interior accuracy. NOVA decomposes neural basis coefficients into a particular solution and a nullspace component. This enforces hard boundary constraints without penalty tuning. To reduce full SVD costs, we derive a horizontal nullspace transport equation. We justify this equation using the Stiefel principal

bundle over the Grassmannian. Based on these developments, we design an efficient algorithm. NOVA achieves SOTA boundary accuracy while maintaining high interior accuracy on the evaluated multiphysics benchmarks.

My Contributions:

- Designed the NOVA framework to enforce hard boundary constraints through nullspace decomposition. Independently completed all theoretical derivations, establishing the mathematical foundation for nullspace transport.
- Designed the transport algorithm based on these derivations to reduce full SVD costs. Independently implemented NOVA, baseline methods, and all experimental code. Organized the complete codebase.
- Conducted the literature review. Designed all benchmarks, baseline comparisons, and ablation studies. Independently ran every experiment. Wrote the full manuscript, covering the method, theoretical analysis, and experimental findings.

OrthVMS Orthogonal Variational Multiscale Correction for Full-Rank Updates with Low-Rank Optimizer States

Submitted to ICLR 2027

Zhuoli Ouyang (Project Leader), Yujun He, Hongbin Zhong, et al.

Advisor: Prof. Zhihai He

Abstract: LLM training requires substantial GPU memory. Existing low-rank optimizers save memory but underuse gradient information in the orthogonal complement. We propose OrthVMS, a memory-efficient optimizer. Inspired by Variational Multiscale (VMS) in computational mechanics, we treat the projected space as coarse and its orthogonal complement as fine. We construct a full-space preconditioner from decoupled coarse and fine preconditioners. OrthVMS achieves SOTA performance among memory-efficient optimizers on diverse pretraining benchmarks.

My Contributions:

- Introduced Variational Multiscale (VMS) theory from computational mechanics into LLM optimizer design. Treated the compact subspace as coarse and its orthogonal complement as fine to construct a full-space preconditioner.
- Developed the complete convergence analysis. Laid the theoretical foundation for the paper. Designed the novel algorithm based on the theoretical derivations. Independently implemented OrthVMS, baselines, and the evaluation framework.
- Conducted the literature review. Designed pretraining experiments with progressively increasing sequence lengths. Developed the paper’s narrative and structure. Organized the content and wrote the full manuscript.

Publications

NOVA Nullspace Dynamics in Neural Basis Solvers for Moving Boundary PDEs, Transport, Rank Transitions, and Reprojection	ICLR 2027 (Submitted)
Zhuoli Ouyang, Changxi Chi, Siyuan Li, Tailin Wu	
OrthVMS Orthogonal Variational Multiscale Correction for Full-Rank Updates with Low-Rank Optimizer States	ICLR 2027 (Submitted)
Zhuoli Ouyang, Yujun He, Hongbin Zhong, et al.	
ENGINE Endogenous Variational MultiScale Optimization for Zeroth-Order LLM Fine-Tuning	NeurIPS 2026 (Poster)
Zhuoli Ouyang, Changxi Chi, Siyuan Li, Tailin Wu	
RMNP Row-Momentum Normalized Preconditioning for Scalable Matrix-Based Optimization	ICML 2026 (Regular)
Shenyang Deng*, Zhuoli Ouyang*, Tianyu Pang, et al. *Equal contribution	
Understanding Generalization from Embedding Dimension and Distributional Convergence	ICML 2026 (Regular)
Junjie Yu*, Zhuoli Ouyang*, Haotian Deng*, et al. *Equal contribution	
Depth, Not Data An Analysis of Hessian Spectral Bifurcation	ISIT 2026
Shenyang Deng*, Boyao Liao*, Zhuoli Ouyang*, et al. *Equal contribution	
Suspicious Alignment of SGD A Fine-Grained Step Size Condition Analysis	ALT 2026
Shenyang Deng, Boyao Liao, Zhuoli Ouyang, et al.	

Full publication list on my homepage

Honors & Awards

- 2026 ALT 2026 Best Student Paper Award
- 2026 ISIT 2026 Strong Accept
- 2026 DATE 2026 Oral Paper
- 2024 SUSTech, Department of Electronics and Electrical Engineering, Honor Class
- 2023 SUSTech, Mathematics Department, Fields Elite Class (Top 1%, Honor Class)

Skills

Programming	Python, C, C++, MATLAB, LaTeX, etc.
Tools & Frameworks	PyTorch, PETSc, Linux, Git, Docker, MATLAB, Fluent
AI Tools	Proficient in using Claude Code, Codex, and AI agents for literature review, method exploration, code implementation, experiment execution, and result analysis.